

Exploring the Potential of Flexible 8-bit Format: Design and Algorithm

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Abstract

Neural network quantization is widely used to reduce model inference complexity in real-world deployments. However, traditional integer quantization suffers from accuracy degradation when adapting to various dynamic ranges. Recent research has focused on a new 8-bit format, FP8, with hardware support for both training and inference of neural networks but lacks guidance for hardware design. In this paper, we analyze the benefits of using FP8 quantization and provide a comprehensive comparison of FP8 with INT quantization. Then we propose a flexible mixed-precision quantization framework that supports various number systems, enabling optimal selection of the most appropriate quantization format for different neural network architectures. Experimental results demonstrate that our proposed framework achieves competitive performance compared to full precision on various tasks, including image classification, object detection, segmentation, and natural language understanding. Our work furnishes critical insights into the tangible benefits and feasibility of employing FP8 quantization, paving the way for heightened neural network efficiency in tangible scenarios. Our code is available in the supplementary material.

1 Introduction

Deep neural networks (DNNs) have achieved significant achievements across domains like computer vision, and natural language processing. Nonetheless, their deployment on resource-limited devices poses challenges owing to their computational and memory intensiveness. Several strategies, such as knowledge distillation, sparsity, and quantization, have been developed to mitigate these challenges.

Among these strategies, quantization is a widely adopted technique that involves reducing the precision of the model's weights and activations. The popularity of INT8 quantization (Jacob et al. 2018a; Li et al. 2021b,a; Nagel et al. 2020; Zhu et al. 2020) stems from its ability to deliver impressive performance with minimal accuracy reduction. However, for specific tasks like semantic segmentation and object detection, INT8 quantization can notably impair accuracy (Jouppi et al. 2021). This has led to the emergence of new data formats, notably BF16 and FP8.

Although hardware support for FP8 (Micikevicius et al. 2022) is primarily centered around training DNNs, compre-

hensive exploration, and analysis of FP8 remain in their infancy. While (Huang, Chen, and Huang 2021) did introduce a flexible 8-bit floating-point format, it lacked universality across networks. There's a conspicuous absence of systematic insights regarding the suitability of FP8 or INT8 quantization for various scenarios, and guidance for hardware design is also scarce.

Addressing these gaps, our study provides an in-depth analysis of the pros and cons of diverse data formats and standards. We present a flexible 8-bit design optimized for speed and precision, supporting a blend of different number systems INT and FP to cater to varying data distributions. Importantly, we delineate the data flow of these flexible formats, offering invaluable guidance for hardware design. We've also crafted a highly efficient automatic mixed-precision search algorithm to pinpoint the ideal format combination tailored to distinct network architectures, with E3M4 emerging as dominant across our DNN experiments. Our method's efficacy is underscored by its performance across tasks such as image classification, object detection, semantic segmentation, text-to-image generation, and natural language understanding (NLU), achieving a full-precision level.

To summarize, our contributions are as follows:

1. We introduce and select a flexible 8-bit format that is friendly to both speed and precision. Our format supports a mixture of different number systems, including INT and FP, to cope with different data distributions.
2. We provide an exhaustive outline of the data flow of flexible formats, detailing aspects like data storage and computational processes, thereby steering hardware design toward efficiency.
3. We have designed an efficient automatic mixed-precision search algorithm to explore the optimal format combination for different network architectures and tasks. Our algorithm can quickly search for the best format for a given network architecture, achieving high performance with low computational cost.
4. We demonstrate the effectiveness of our proposed method on various tasks, including image classification, object detection, semantic segmentation, text-to-image generation, and NLU. Our method achieves full-precision level accuracy while being highly efficient and

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